Project Documentation

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MCD | MAESTRÍA EN CIENCIA DE LOS DATOS

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1.1 Dataset Extraction

The data used in this project consists of Glassdoor employee reviews.

Source: The dataset is stored in a structured CSV format within the data/ directory.

Loading Process: Data ingestion is performed using the pandas library. The script dynamically locates the file using absolute paths to ensure portability across different environments.

Format: The primary features extracted are the Pros (textual reviews), which are used for sentiment analysis and language detection.

3.2 Model Construction

The architecture follows a modular design to separate concerns and improve maintainability:

Preprocessing Module (preprocessing.py): Implements text normalization, including noise removal (special characters, extra spaces) and formatting for NLP tasks.

Sentiment Analysis Module (model_training.py): * Bilingual Support: Uses langdetect to identify the review's language.

Hybrid Approach: Utilizes VADER for English texts (optimized for social media/short reviews) and TextBlob for Spanish texts, ensuring high accuracy across different languages.

Logic Flow: The model receives raw text, detects the language, cleans the content, and assigns a sentiment score (Positive, Neutral, or Negative).

3.3 MLOps Integration

The project integrates MLflow to manage the machine learning lifecycle and ensure experiment traceability:

Tracking Server: All runs are logged to a local MLflow server (http://127.0.0.1:5000).

Experiment Management: Each execution is recorded under the "Glassdoor_Analysis" experiment.

Logged Metadata: * Metrics: Tracks total_reviews processed in each run.

Artifacts: Automatically generates and stores a visualization report (reporte_sentimiento.png) for every execution.

Reproducibility: By wrapping the pipeline in mlflow.start_run(), we guarantee that every change in the preprocessing or model logic is documented.

3.4 Execution Guide

To run this project locally, follow these steps:

Environment Setup: Ensure you have the required libraries installed (pandas, mlflow, langdetect, nltk, textblob).

Start MLflow Server: Open your terminal and run:

Bash

```
mlflow ui
```

Run the Pipeline:

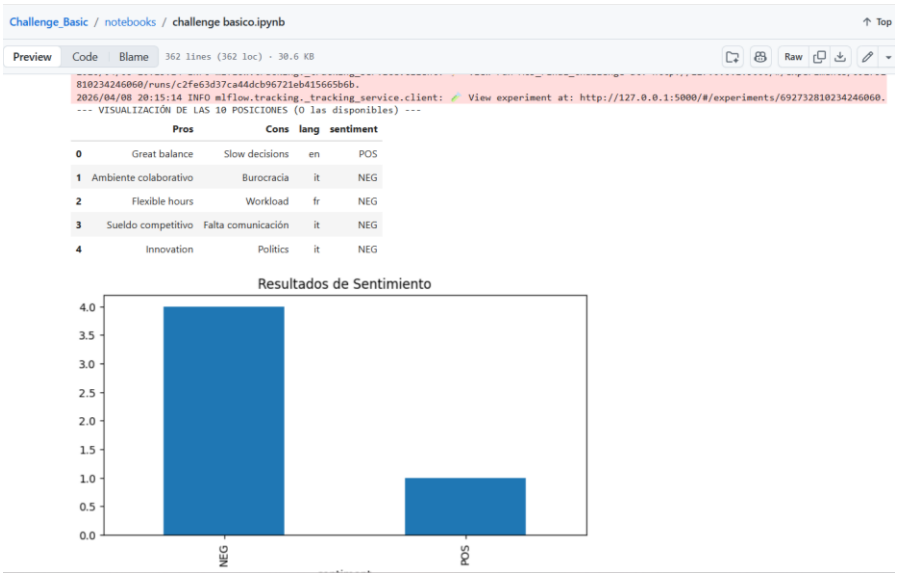
Open the Jupyter Notebook in the notebooks/ folder.

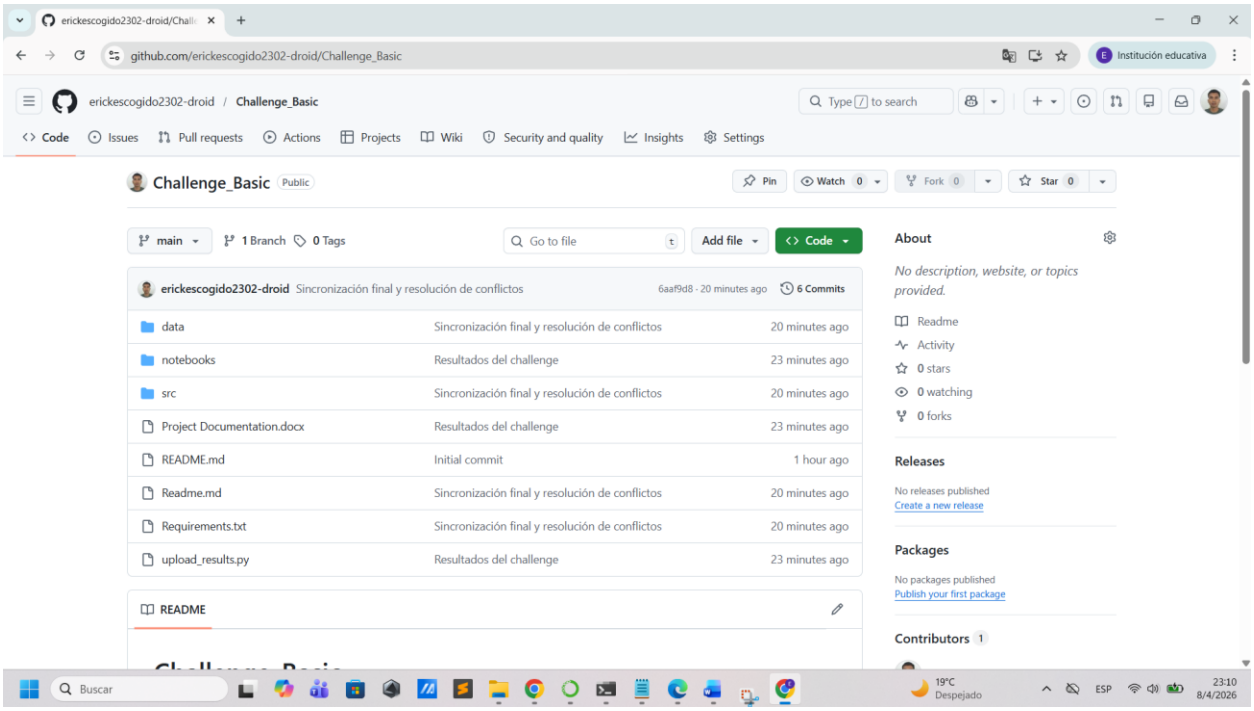
Execute the cells to load the modules from the src/ directory.

Call the run_mlops_pipeline(df) function.

View Results: Access the MLflow Dashboard at http://localhost:5000 to inspect metrics and download the sentiment report.

Evidence.

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notebooks Resultados del challenge 23 minutes ago
src Sincronización final y resolución de conflictos 20 minutes ago
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Readme.md Sincronización final y resolución de conflictos 20 minutes ago
Requirements.txt Sincronización final y resolución de conflictos 20 minutes ago
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About
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Activity
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0 watching
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No releases published
Create a new release
Packages
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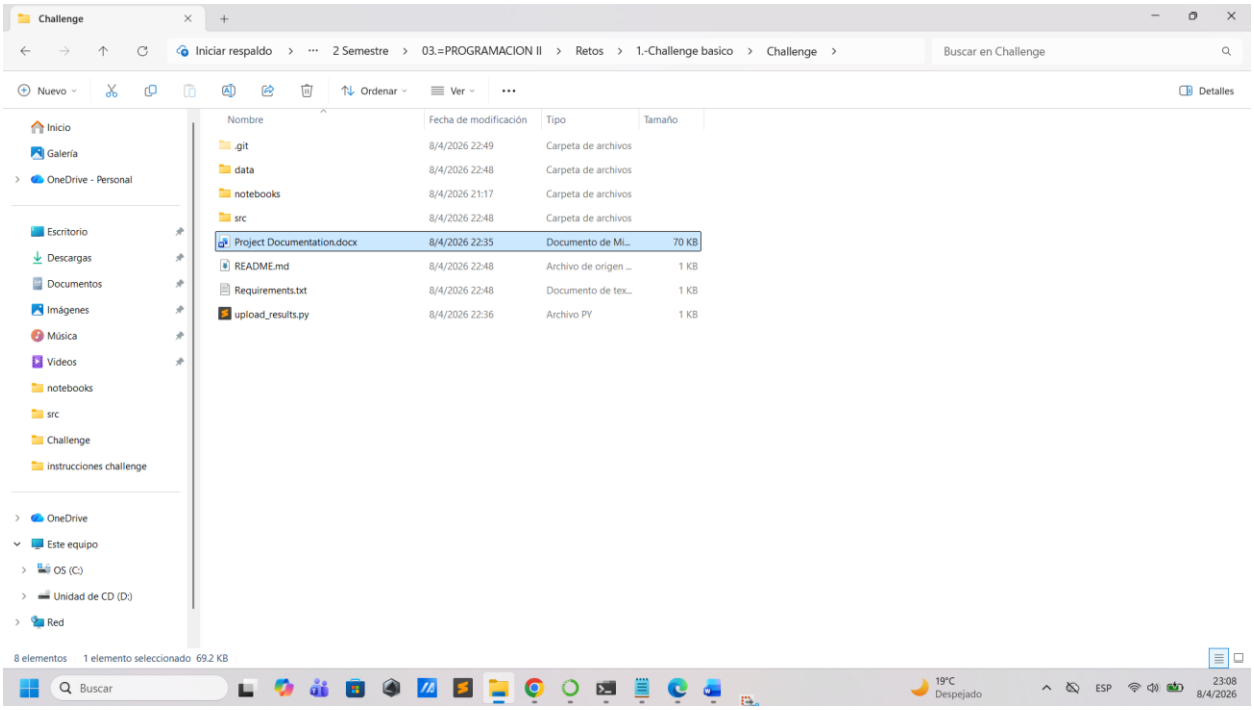
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